

Food Science Lessons





About the Course

An elective in Physical Science, Food Science introduces learners to applications in chemistry and physics as they relate to cooking and baking, while incorporating historical context, modern developments, current events, and citizenship. This is a hands-on course that requires independent interest and some flexibility for work time in the kitchen. There is room for teachers and learners to adjust the course for personal needs and preferences.

This topic is included in the following course(s): Science: Food Science



Placement & Combining Tips

There are no specific prerequisites or concurrents for this course.



Scheduling

GRADE	SCHEDULE INFO.	BOOKS
9-12	Food Science Lessons 5 times/week 45 min	Salt, Fat, Acid, Heat: Mastering the Elements of Good Cooking The Elements of Baking Gullah Geechee Home Cooking: Recipes from the Matriarch of Edisto Island Mexico in Your Kitchen Classic Russian Cooking Pomp and Sustenance Celtic Folklore Cooking Culinary Reactions: The Everyday Chemistry of Cooking

Sample Weekly View

Day 1	Day 2	Day 3	Day 4	Day 5
Science: Food Science				
Food Science Lessons Nature Walks & Scouting: Grades 9-12	Food Science Lessons	Food Science Lessons	Food Science Lessons Nature Notebook: Grades 9-12	Food Science Lessons Food Science Labs



Planning & Prep

Permission to print for non-commercial use. See Alveary group use policy to use lessons in a group context.

LINKS: Click text or scan the QR code in the top corner of the lesson plan pages to view online resources associated with the lessons.

Responsibility for previewing all links rests with the teacher. All links were checked at the time of publication; however, websites change frequently and may contain objectionable content. Please report broken links by contacting us through our website.

Food Science Lessons

☐ Carve out time to continue or establish the regular habit of spending time in nature, including the use of a nature journal, as appropriate.

- ☐ Obtain materials from the supply lists.
- ☐ The lab period is the student's opportunity to prepare a meal (or part of a meal, depending on teacher preference). Therefore, it should be scheduled at a time when there will be people to eat what they prepare, which may not be during their school day.
- ☐ Select a science book from the Alveary bookshelf for personal reading time, as appropriate.
- ☐ Bookmark or print Quick Links, as needed.

Term Prep & Teacher Tips

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- ☐ Gather household items, typically easy for students to scavenge or teachers to obtain locally:
 - basic cookware and bakeware (pots, pans, baking sheets, mixing bowls)
 - basic cooking utensils (mixing spoons, spatula, tongs, knives, measuring cups and spoons, whisk, meat thermometer)
 - kitchen scale
 - mortar and pestle
 - salt and pepper
 - romaine lettuce (Term 1 Week 1)
 - anchovies (Term 1 Week 1)
 - mayonnaise or an egg to make fresh (Term 1 Week 1)
 - garlic clove (Term 1 Week 1)
 - lemon juice (Term 1 Week 1)
 - white wine vinegar (Term 1 Week 1)
 - parmesan cheese (Term 1 Week 1)
 - Worcestershire sauce (Term 1 Week 1)
 - olive oil (Term 1 Week 1)
 - bread of your choice (Term 1 Week 1)
 - dinner protein of your choice (Term 1 Week 1)
 - ingredients by student choice (students are prompted to make a grocery list most weeks)
 - optional: additional kitchen tools as indicated in cookbooks, depending on recipes chosen by students

Reminders

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- ☐ Note that Lab in week 1 calls for lettuce, anchovies, mayonnaise/egg, garlic clove, lemon juice, white wine vinegar, parmesan cheese, Worcestershire sauce, olive oil, and bread and dinner protein of choice. Additional grocery lists will be provided by the student throughout the year.



Books & Resources

For book rationales and purchase options, click the Book List link or scan the QR code below.

∞ [View Book List Details](#)

Food Science Lessons



Salt, Fat, Acid, Heat: Mastering the Elements of Good Cooking



The Elements of Baking



Gullah Geechee Home Cooking: Recipes from the Matriarch of Edisto Island



Mexico in Your Kitchen



Classic Russian Cooking



Pomp and Sustenance



Celtic Folklore Cooking



Culinary Reactions: The Everyday Chemistry of Cooking



Supplies

For supply list details and basic supplies helpful to have on hand, click the links or scan the QR code below.

∞ [View Basic Supplies](#)

(No Subject Supplies Assigned)



Quick Links

Food Science Lessons

- ∞ [Extra Helpings](#)
- ∞ [America's Test Kitchen Library](#)
- ∞ [The Loopy Whisk](#)
- ∞ [Samin Nosrat Cooking](#)
- ∞ [Food & Wine](#)
- ∞ [Science News Explores](#)

Click THIS text
or scan the QR
code for links.



Food Science Lessons

How To Approach



Introduce

- If starting a new book or a new topic in the book, then look at the title or a picture and take a moment to consider previous ideas and experiences.
- If continuing a previous reading, recap what was read previously. Often, the title of the book's section can help to draw out the main idea.



Read

- Read or do, as instructed in the lessons, making note of or flagging unfamiliar terms, interesting ideas, important dates, inspiring quotes, etc.
- Use supportive strategies and educational tools to reduce frustration and better engage the mind, as appropriate. These could include, but are not limited to, the use of eBooks, pictures, audio, read-aloud, buddy reading, colored reading strips, etc.
- As they read, learners record ideas in a notebook or binder using outlines, diagrams, graphic organizers, or other methods (or a combination of methods) that suit them. These recordings can be a helpful mechanism for remembering or a mini-narration to support understanding.
- If learners do not understand a word or concept, do not worry. Try reading over the passage again, studying a picture or diagram, connecting the idea to something from real life, or practicing chapter exercises. The lab/field work further supports major concepts from the text.



Narrate

- Process the ideas of the lesson by retelling, defining a concept, explaining the links in a chain of thought, etc. Do this orally or silently to yourself.
- Use words, pictures, outlines, etc.
- If a particular idea cannot be narrated, then read or examine the text again.



Discuss

- Consider with the teacher any thoughts, confusion, or concerns about the passage.
- If understanding is still uncertain, try rereading the passage or do some personal research on the topic.



Connect

- Follow any extra links, examine any sidebars in the text, or pursue additional reading, depending on student interest.

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[Click THIS text or scan the QR code for links.](#)



Term 1

WEEK 1 45m Food Science Lessons - Lesson 1

Preparation

Materials: Video Link

→ INTRO

Your lessons are organized to teach you the basics in Term 1 and then encourage exploration in Terms 2-3. If you prefer to spread out the readings in Term 1, there is room to do so. Feel free to work at your own pace.

As we start our kitchen adventure, it is important to remember that we are also investigating as scientists. Let's watch a portion of this public lecture from the Science & Cooking lecture series at Harvard to get us in the mindset that the best chefs must be scientists, too.

→ VIEW, NARRATE, & DISCUSS

Watch a portion of the first Science and Cooking lecture:

∞ Video Link: Science and Cooking (5:45-57:14)

• PLAN WEEKLY

- take a nature walk
- science free read

WEEK 1 45m Food Science Lessons - Lesson 2

The Science of Cooking

Materials: Salt, Fat, Acid, Heat

→ INTRO

In this book, a chef leads students through 4 important "elements" of cooking: salt, fat, acid, and heat. The first 3 are families of chemical substances, while the last is really about adding energy. If you have already taken Chemistry, think about what you've already learned about salt, fat, acid, and heat as you observe the roles they play in cooking. If you haven't yet taken Chemistry, you'll learn a few basics here and will get to make lots of observations that will support any future study in the subject.

→ READ, NARRATE, & DISCUSS

p.17-28

→ VIEW

∞ Video Link: America's Test Kitchen

∞ Video Link: Alveary Chemistry Companion

WEEK 1 45m Food Science Lessons - Lesson 3

The Science of Cooking

Materials: Salt, Fat, Acid, Heat

→ READ, NARRATE, & DISCUSS

p.29-39

WEEK 1 45m Food Science Lessons - Lesson 4

The Science of Cooking

Materials: Salt, Fat, Acid, Heat

→ READ, NARRATE, & DISCUSS

p.40-55



Term 1

WEEK 1 45m Food Science Lessons - Lesson 5

The Science of Cooking & Baking

Materials: Salt, Fat, Acid, Heat, Elements of Baking

→ PREPARE

Choose a 'Salt Lesson' recipe from p.428 of SFAH for next week. Then decide what else you will serve to balance the meal. Healthy Eating Plate (linked) may help; for more information about the history of dietary guidelines, the newly released U.S. guidelines, or Canadian guidelines, check out the other optional links. Be sure to make a shopping list for your teacher.

∞ Website Link: Healthy Eating Plate

∞ Website Link: Illustrating Diet Advice is Hard

∞ Website Link: Dietary Guidelines for Americans

∞ Website Link: Canada's Dietary Guidelines

Then begin Elements of Baking:

→ INTRO

In the second text for this course, a chemist leads students through studies in baking with the mindset of a materials chemist: we observe the physical properties of various materials and experiment with how they perform in various situations to work toward a desired product. The approach is the same whether we are developing new adhesives for space shuttle tiles, new polymers for biomedical applications or textiles, or new recipes for friends and family.

Note the organization of EB on p.13. Feel free to flip around so that you understand how the book is organized and where to find what you want.

Read p.15. Note that you will measure by weight in this course. In science, we experiment, and the accuracy and precision of weight measurements will ensure that we can draw conclusions from our experiments.

→ READ, NARRATE, & DISCUSS

p.19-23

WEEK 2 45m Food Science Lessons - Lesson 6

The Science of Baking

Materials: Elements of Baking

→ READ, NARRATE, & DISCUSS

Ch.2 p.37-44

• PLAN WEEKLY

- ☐ take a nature walk
- ☐ science free read

WEEK 2 45m Food Science Lessons - Lesson 7

The Science of Baking

Materials: Elements of Baking

→ READ, NARRATE, & DISCUSS

Ch.3 p.48-49. Read once before case studies, but you will come back to them after (or even during) case studies as you develop understanding. Make sure you understand WHY the points on p.49 are so.

→ PLAN

Look over p.50-51: In addition to Buttery Vanilla Cake, select 3 items from the pink boxes which you are familiar with or interested in (except bread, which we'll get to later, and waffles).

→ PREPARE

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Term 1

Read over the rules for Buttery Vanilla Cake (p.52-54) and your 3 additional items, noting how the rules apply the science of ingredients from Ch.2, how the vegan and gf-vegan rules combine the first 3, and how the rules are the same/different between items.

Refer to p.91 for a Table of Contents of case studies to locate the recipe pages for your 3 selections. Note that a table of the ingredients needed for different variations is printed toward the end of the case study (p.104 for Buttery Vanilla Cake). You can choose whatever variations most interest you, but the key to learning the science is to try different variations because we have to change a variable to see what happens. Add any needed ingredients to your shopping list.

WEEK 2 45m Food Science Lessons - Lesson 8

The Science of Baking

Materials: Elements of Baking

→ READ, NARRATE, & DISCUSS

Ch.4 p.88-89

Buttery Vanilla Cake p.92-105. Even if you aren't planning to do all of the variations, reading about the role the ingredients play is important!

WEEK 2 45m Food Science Lessons - Lesson 9

The Science of Baking

Materials: Elements of Baking, indicated ingredients, and kitchen supplies

→ BAKING DAY!

Prepare Buttery Vanilla Cake, recording what you do and what you observe in your notebook.

WEEK 2 45m Food Science Lessons - Lesson 10

The Science of Baking

Materials: Elements of Baking

→ ANALYZE

- Review how the bake went, reread, and plan any adjustments that you want to try next week.
- Add frostings and creams from Ch.10, if desired.
- Add to shopping list, as needed.
- If you want to decide on your Salt Lesson recipe for next week, you could do this now, or you could wait until after dinner night. You could try one of these first two again or try a new one. Add to shopping list, as needed.

WEEK 3 45m Food Science Lessons - Lesson 11

The Science of Baking

Materials: Elements of Baking

→ READ, NARRATE, & DISCUSS

Ch.5 p.259-266, p.267-268 (optional)

• PLAN WEEKLY

- ☐ take a nature walk
- ☐ science free read

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[Click THIS text or scan the QR code for links.](#)



Term 1

WEEK 3 45m Food Science Lessons - Lesson 12

The Science of Baking

Materials: Elements of Baking

→ READ, NARRATE, & DISCUSS

Ch.5 p.269-275

WEEK 3 45m Food Science Lessons - Lesson 13

The Science of Baking

Materials: Elements of Baking, indicated ingredients, and kitchen supplies

→ BAKING DAY!

Prepare Buttery Vanilla Cake, including any changes or adjustments noted last week. Record what you do and what you observe in your notebook.

WEEK 3 45m Food Science Lessons - Lesson 14

The Science of Baking

Materials: Elements of Baking

→ ANALYZE

- Review how the bake went, reread, and plan any adjustments that you want to try next week OR
- Select your next case study by reading any introduction, as well as the recipe (you will have more time to finish reading later, but don't wait to add to the shopping list, as needed). Even though you have read about bread in the gluten-free chapter, complete your case studies before moving on to bread, as this brings in new material to consider (psyllium husk).

WEEK 3 45m Food Science Lessons - Lesson 15

The Science of Cooking

Materials: Salt, Fat, Acid, Heat

→ READ, NARRATE, & DISCUSS

p.205, 208-223

→ PLAN

Consider your Salt Lesson for this week. Will you use any of the ideas in this reading for the recipe? If not, you can always come back to them in a future recipe.

If you want to decide on your Salt Lesson recipe for next week, you could do this now, or you could wait until after dinner night. You could try one of these first two again or try a new one. Keep in mind that any salad can be a Salt Lesson, so feel free to experiment with the Ideal Salad (p.216) or Avocado Matrix (p.222). Add to shopping list as needed.

WEEK 4 45m Food Science Lessons - Lesson 16

The Science of Baking

Materials: Elements of Baking

→ READ, NARRATE, & DISCUSS

Finish any reading from last week, then read Ch.6 p.311-317.

• PLAN WEEKLY

- ☐ take a nature walk
- ☐ science free read

Food Science Lessons

[Click THIS text or scan the QR code for links.](#)



Term 1

→ PREPARE

Complete any preparation that you need to do for your bake tomorrow.

WEEK 4 45m Food Science Lessons - Lesson 17

Materials: Elements of Baking, indicated ingredients, and kitchen supplies

→ BAKING DAY!

Prepare Buttery Vanilla Cake, including any changes or adjustments noted last week. Record what you do and what you observe in your notebook.

WEEK 4 45m Food Science Lessons - Lesson 18

The Science of Baking

Materials: Elements of Baking

→ ANALYZE

- Review how the bake went, reread, and plan any adjustments that you want to try next week OR
- Select your next case study by reading any introduction, as well as the recipe. Add to the shopping list, as needed.

WEEK 4 45m Food Science Lessons - Lesson 19

The Science of Cooking

Materials: Salt, Fat, Acid, Heat

→ READ, NARRATE, & DISCUSS

p.57-71

→ VIEW

- ∞ Video Link: America's Test Kitchen
- ∞ Video Link: Alveary Chemistry Companion

WEEK 4 45m Food Science Lessons - Lesson 20

The Science of Cooking

Materials: Salt, Fat, Acid, Heat

→ READ, NARRATE, & DISCUSS

p.72-87

→ PLAN

Consider your Salt Lesson for this week. Will you use any of the ideas in this reading for the recipe? If not, you can always come back to them in a future recipe.

If you want to decide on your Cooking Lesson recipe for next week, you could do another Salt Lesson or try a Fat Lesson (p.429). You could wait to decide until after dinner night, too. Add to shopping list, as needed.

WEEK 5 45m Food Science Lessons - Lesson 21

The Science of Cooking

Materials: Salt, Fat, Acid, Heat

→ READ, NARRATE, & DISCUSS

p.88-99

• PLAN WEEKLY

- take a nature walk
- science free read

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Term 1

WEEK 5 45m Food Science Lessons - Lesson 22

The Science of Baking

Materials: Elements of Baking, indicated ingredients, and kitchen supplies

→ BAKING DAY!

Prepare your selected recipe. Record what you do and what you observe in your notebook.

WEEK 5 45m Food Science Lessons - Lesson 23

The Science of Baking

Materials: Elements of Baking

→ ANALYZE

- Review how the bake went, reread, and plan any adjustments that you want to try next week OR
- Select your next case study by reading any introduction, as well as the recipe. Add to the shopping list, as needed.

WEEK 5 45m Food Science Lessons - Lesson 24

The Science of Baking

Materials: Elements of Baking

→ READ, NARRATE, & DISCUSS

Ch.7 p.343-349

WEEK 5 45m Food Science Lessons - Lesson 25

The Science of Baking

Materials: Elements of Baking

→ READ, NARRATE, & DISCUSS

Ch.7 p.350-355. Compare the application here, but don't worry about mastering or remembering it. You can always come back as you try particular recipes.

→ PLAN

Decide on your Cooking Lesson recipe for next week, or wait to decide until after dinner night. Add to shopping list, as needed.

WEEK 6 45m Food Science Lessons - Lesson 26

The Science of Baking

Materials: Elements of Baking

→ READ, NARRATE, & DISCUSS

Ch.7 p.356-360

• PLAN WEEKLY

- ☐ take a nature walk
- ☐ science free read

WEEK 6 45m Food Science Lessons - Lesson 27

The Science of Baking

Materials: Elements of Baking, indicated ingredients, and kitchen supplies

→ BAKING DAY!

Prepare your selected recipe. Record what you do and what you observe in your notebook.

Food Science Lessons

[Click THIS text or scan the QR code for links.](#)



Term 1

WEEK 6 ☐ 45m Food Science Lessons - Lesson 28

The Science of Baking

☐ Materials: Elements of Baking

→ ANALYZE

- Review how the bake went, reread, and plan any adjustments that you want to try next week OR
- Select your next case study by reading any introduction, as well as the recipe. Add to the shopping list, as needed.

WEEK 6 ☐ 45m Food Science Lessons - Lesson 29

The Science of Baking

☐ Materials: Elements of Baking

→ READ, NARRATE, & DISCUSS

Ch.8 p.393-398

WEEK 6 ☐ 45m Food Science Lessons - Lesson 30

The Science of Baking

☐ Materials: Elements of Baking

→ REREAD & DISCUSS

Ch.3. Check that the ideas make sense now that you have some experience.

→ PLAN

Decide on your Cooking Lesson recipe for next week, or wait to decide until after dinner night. Add to shopping list, as needed.

WEEK 7 ☐ 45m Food Science Lessons - Lesson 31

The Science of Baking

☐ Materials: Elements of Baking

→ READ, NARRATE, & DISCUSS

Ch.9 p.427-433

• PLAN WEEKLY

- ☐ take a nature walk
- ☐ science free read

WEEK 7 ☐ 45m Food Science Lessons - Lesson 32

The Science of Baking

☐ Materials: Elements of Baking, indicated ingredients, and kitchen supplies

→ BAKING DAY!

Prepare your selected recipe. Record what you do and what you observe in your notebook.

WEEK 7 ☐ 45m Food Science Lessons - Lesson 33

The Science of Baking

☐ Materials: Elements of Baking

→ ANALYZE

- Review how the bake went, reread, and plan any adjustments that you want to try next week OR
- Select your next case study by reading any introduction, as well as the recipe. Add to the shopping list, as needed.

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Term 1

WEEK 7 45m Food Science Lessons - Lesson 34

The Science of Cooking

Materials: Salt, Fat, Acid, Heat

→ READ, NARRATE, & DISCUSS

p.101-110

→ VIEW

∞ Video Link: America's Test Kitchen

∞ Video Link: Alveary Chemistry Companion

WEEK 7 45m Food Science Lessons - Lesson 35

The Science of Cooking

Materials: Salt, Fat, Acid, Heat

→ READ, NARRATE, & DISCUSS

p.111-119

→ PLAN

Decide on your Cooking Lesson recipe for next week, or wait to decide until after dinner night. Add to shopping list, as needed.

WEEK 8 45m Food Science Lessons - Lesson 36

The Science of Cooking

Materials: Salt, Fat, Acid, Heat

→ READ, NARRATE, & DISCUSS

p.120-129

• PLAN WEEKLY

take a nature walk

science free read

WEEK 8 45m Food Science Lessons - Lesson 37

The Science of Baking

Materials: Elements of Baking, indicated ingredients, and kitchen supplies

→ BAKING DAY!

Prepare your selected recipe. Record what you do and what you observe in your notebook.

WEEK 8 45m Food Science Lessons - Lesson 38

The Science of Baking

Materials: Elements of Baking

→ ANALYZE

• Review how the bake went, reread, and plan any adjustments that you want to try next week OR

• Select your next case study by reading any introduction, as well as the recipe. Add to the shopping list, as needed.

WEEK 8 45m Food Science Lessons - Lesson 39

The Science of Cooking

Materials: Salt, Fat, Acid, Heat

→ READ, NARRATE, & DISCUSS

p.238-239

→ ANALYZE

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Term 1

Spend some time comparing different recipes in SFAH to see how Salt, Fat, and Acid work together before we examine Heat. Rather than another material ingredient, Heat stands apart because what we are adding is energy.

WEEK 8 45m Food Science Lessons - Lesson 40

The Science of Cooking

Materials: Salt, Fat, Acid, Heat

→ ANALYZE

Compare different recipes in SFAH to see how Salt, Fat, and Acid work together.

→ PLAN

Decide on your Cooking Lesson recipe for next week, or wait to decide until after dinner night. Add to shopping list, as needed.

WEEK 9 45m Food Science Lessons - Lesson 41

The Science of Cooking

Materials: Salt, Fat, Acid, Heat

→ READ, NARRATE, & DISCUSS

p.131-139

→ VIEW

∞ Video Link: Alveary Chemistry Companion

• PLAN WEEKLY

- take a nature walk
- science free read

WEEK 9 45m Food Science Lessons - Lesson 42

The Science of Baking

Materials: Elements of Baking, indicated ingredients, and kitchen supplies

→ BAKING DAY!

Prepare your selected recipe. Record what you do and what you observe in your notebook.

WEEK 9 45m Food Science Lessons - Lesson 43

The Science of Baking

Materials: Elements of Baking

→ ANALYZE

- Review how the bake went, reread, and plan any adjustments that you want to try next week OR
- Select your next case study by reading any introduction, as well as the recipe. Add to the shopping list, as needed.

WEEK 9 45m Food Science Lessons - Lesson 44

The Science of Cooking

Materials: Salt, Fat, Acid, Heat

→ READ, NARRATE, & DISCUSS

p.140-150

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[Click THIS text or scan the QR code for links.](#)



Term 1

WEEK 9 ☐ 45m Food Science Lessons - Lesson 45

The Science of Cooking

☐ Materials: Salt, Fat, Acid, Heat

→ READ, NARRATE, & DISCUSS

p.151-160

→ PLAN

Decide on your Cooking Lesson recipe for next week, or wait to decide until after dinner night. Add to shopping list, as needed.

WEEK 10 ☐ 45m Food Science Lessons - Lesson 46

The Science of Cooking

☐ Materials: Salt, Fat, Acid, Heat

→ READ, NARRATE, & DISCUSS

p.161-170 (end of Reducing)

• PLAN WEEKLY

☐ take a nature walk

☐ science free read

WEEK 10 ☐ 45m Food Science Lessons - Lesson 47

The Science of Baking

☐ Materials: Elements of Baking, indicated ingredients, and kitchen supplies

→ BAKING DAY!

Prepare your selected recipe. Record what you do and what you observe in your notebook.

WEEK 10 ☐ 45m Food Science Lessons - Lesson 48

The Science of Baking

☐ Materials: Elements of Baking

→ ANALYZE

• Review how the bake went, reread, and plan any adjustments that you want to try next week OR

• Select your next case study by reading any introduction, as well as the recipe. Add to the shopping list, as needed.

WEEK 10 ☐ 45m Food Science Lessons - Lesson 49

The Science of Cooking

☐ Materials: Salt, Fat, Acid, Heat

→ READ, NARRATE, & DISCUSS

p.170-180 (end of Baking)

WEEK 10 ☐ 45m Food Science Lessons - Lesson 50

The Science of Cooking

☐ Materials: Salt, Fat, Acid, Heat

→ READ, NARRATE, & DISCUSS

p.180-190

→ PLAN

Decide on your Cooking Lesson recipe for next week, or wait to decide until after dinner night. Add to shopping list, as needed.

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Term 1

WEEK 11 45m Food Science Lessons - Lesson 51

The Science of Cooking

Materials: Salt, Fat, Acid, Heat

→ **READ, NARRATE, & DISCUSS**

p.191-200

• PLAN WEEKLY

- ☐ take a nature walk
- ☐ science free read

WEEK 11 45m Food Science Lessons - Lesson 52

The Science of Baking

Materials: Elements of Baking, indicated ingredients, and kitchen supplies

→ **BAKING DAY!**

Prepare your selected recipe. Record what you do and what you observe in your notebook.

WEEK 11 45m Food Science Lessons - Lesson 53

The Science of Baking

Materials: Elements of Baking

→ **ANALYZE**

- Review how the bake went and consider what you learned.

WEEK 11 45m Food Science Lessons - Lesson 54

Present What You've Learned

→ **PREPARE**

For one of your exam questions, you will be asked to give an informal presentation to your teacher on something you have learned about during lessons this term, such as how a particular recipe or ingredient works. (This is distinct from what you learned by preparing meals on dinner night.) Use this time to gather your thoughts and to think through how you want to do that. You can be as creative as you would like. If you want to create something to present, use this time to work on it.

WEEK 11 45m Food Science Lessons - Lesson 55

Present What You've Learned

→ **PREPARE**

For one of your exam questions, you will be asked to give an informal presentation to your teacher on something you have learned about during lessons this term, such as how a particular recipe or ingredient works. (This is distinct from what you learned by preparing meals on dinner night.) Use this time to gather your thoughts and to think through how you want to do that. You can be as creative as you would like. If you want to create something to present, use this time to work on it.

WEEK 12 45m Food Science Lessons - Lesson 56

Exam Week

→ **EXAMS**

Answer question(s) related to the course.

WEEK 12 45m Food Science Lessons - Lesson 57

Exam Week

→ **EXAMS**

Answer question(s) related to the course.

Food Science Lessons

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Term 1

WEEK 12  45m Food Science Lessons - Lesson 58 <i>Exam Week</i> → EXAMS Answer question(s) related to the course.	
WEEK 12  45m Food Science Lessons - Lesson 59 <i>Exam Week</i> → EXAMS Answer question(s) related to the course.	
WEEK 12  45m Food Science Lessons - Lesson 60 <i>Exam Week</i> → EXAMS Answer question(s) related to the course.	

Food Science Lessons

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Term 2

WEEK 1 45m Food Science Lessons - Lesson 61

Preparation

Materials: internet access, favorite recipes & cookbooks

→ INTRO

Now it's time to make what you learned in Term 1 entirely your own and apply it more broadly. Terms 2 & 3 will be entirely application. Schedule your baking day and dinner night, but otherwise, spend time comparing recipes, watching any videos (below and in your Quick Links) that you find interesting and helpful, and thinking about how the ideas apply - don't forget to get your shopping list to your teacher on time!

∞ Video Link: America's Test Kitchen has an extensive video library on YouTube (the playlist included in term 1 a few times is called What's Eating Dan)

∞ Video Link: The Loopy Whisk is by the baker and chemist who wrote The Elements of Baking

∞ Video Link: Samin Nosrat Cooking features the chef who wrote Salt Fat Acid Heat

• PLAN WEEKLY

- take a nature walk
- science free read

WEEK 1 45m Food Science Lessons - Lesson 62

The Science of Baking

Materials: Elements of Baking

→ PLAN

Consider what baking lessons you would like to learn this term; what recipes you would like to experiment with (in this book or from any other source); what ingredients you would like to better understand. Set a goal for this term and plan using the skills you have learned.

WEEK 1 45m Food Science Lessons - Lesson 63

The Science of Cooking

Materials: Salt, Fat, Acid, Heat

→ PLAN

Consider what cooking lessons you would like to learn this term; what recipes you would like to experiment with (in this book or from any other source); what ingredients you would like to better understand. Set a goal for this term and plan using the skills you have learned.

WEEK 1 45m Food Science Lessons - Lesson 64

Literature of Food Science & Culture

Materials: culinary anthropology selection

→ INTRO

This term, you will explore how food influences people and how people influence food. You might choose a culture you already know, but would like to better appreciate. You might choose a new culture that you'd like to meet. You might even dabble in more than one culture. The video linked is provided as inspiration:

∞ Video Link: How Pancakes are Made in 11 Countries

→ READ, NARRATE, & DISCUSS

Read at your own pace, as each selection and your interest in them will vary. You could choose to read more during your flex day, during your weekly free read time, or on your own at the end of term. If you choose not to finish the book, that is fine, too. Notice how the ingredients, tools, and customs surrounding food tell a story about the people. Think about how

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the ideas you have learned come together in the cultural development of the people. Use what you have learned to experiment and adopt some new recipes for your family.

WEEK 1 45m Food Science Lessons - Lesson 65

Flex Day

Materials: internet access, favorite recipes & cookbooks, culinary anthropology selection

→ NOTE

Compare recipes, watch any videos (in your Quick Links) that you find interesting and helpful, and think about how the ideas you've learned apply. What do you plan to make next week? Don't forget to get your shopping list to your teacher on time!

OR

Read more from your culinary anthropology selection.

WEEK 2 45m Food Science Lessons - Lesson 66

Preparation

Materials: Elements of Baking, Salt, Fat, Acid, Heat, selected recipes

→ PREPARE

Read over the week's recipes. Consider how the recipes work and any substitutions/adjustments you plan to make. Use any extra time to journal a prelab narration in your notebook or gather ingredients and supplies.

• PLAN WEEKLY

- take a nature walk
- science free read

WEEK 2 45m Food Science Lessons - Lesson 67

The Science of Baking

Materials: Elements of Baking, selected recipe, indicated ingredients, and kitchen supplies

→ BAKING DAY!

Prepare your selected recipe. Record what you do and what you observe in your notebook.

WEEK 2 45m Food Science Lessons - Lesson 68

The Science of Baking

Materials: Elements of Baking, selected recipe

→ ANALYZE

- Review how the bake went, reread, and plan any adjustments that you want to try next week OR
- Select your next recipe. Read the recipe and any relevant information, such as the case study, if the recipe is in Elements of Baking. Add to the shopping list, as needed.

WEEK 2 45m Food Science Lessons - Lesson 69

Literature of Food Science & Culture

Materials: culinary anthropology selection

→ READ, NARRATE, & DISCUSS

Read at your own pace. Notice how the ingredients, tools, and customs surrounding food tell a story about the people. Think about how the ideas

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you have learned come together in the cultural development of the people. Use what you have learned to experiment and adopt some new recipes for your family.

WEEK 2 45m Food Science Lessons - Lesson 70

Flex Day

Materials: internet access, favorite recipes & cookbooks, culinary anthropology selection

→ EXPLORE

Compare recipes, watch any videos (in your Quick Links) that you find interesting and helpful, and think about how the ideas you've learned apply. What do you plan to make next week? Don't forget to get your shopping list to your teacher on time!

OR

→ READ, NARRATE, & DISCUSS

Current Events from the links (below and in your Quick Links):

∞ Website Link: Science News Explores

∞ Website Link: Food & Wine

OR

Read more from your culinary anthropology selection.

WEEK 3 45m Food Science Lessons - Lesson 71

Preparation

Materials: Elements of Baking, Salt, Fat, Acid, Heat, selected recipes

→ PREPARE

Read over the week's recipes. Consider how the recipes work and any substitutions/adjustments you plan to make. Use any extra time to journal a prelab narration in your notebook or gather ingredients and supplies.

• PLAN WEEKLY

take a nature walk

science free read

WEEK 3 45m Food Science Lessons - Lesson 72

The Science of Baking

Materials: Elements of Baking, selected recipe, indicated ingredients, and kitchen supplies

→ BAKING DAY!

Prepare your selected recipe. Record what you do and what you observe in your notebook.

WEEK 3 45m Food Science Lessons - Lesson 73

The Science of Baking

Materials: Elements of Baking, selected recipe

→ ANALYZE

• Review how the bake went, reread, and plan any adjustments that you want to try next week OR

• Select your next recipe. Read the recipe and any relevant information, such as the case study, if the recipe is in Elements of Baking. Add to the shopping list, as needed.

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WEEK 3 45m Food Science Lessons - Lesson 74

Literature of Food Science & Culture

Materials: culinary anthropology selection

→ READ, NARRATE, & DISCUSS

Read at your own pace. Notice how the ingredients, tools, and customs surrounding food tell a story about the people. Think about how the ideas you have learned come together in the cultural development of the people. Use what you have learned to experiment and adopt some new recipes for your family.

WEEK 3 45m Food Science Lessons - Lesson 75

Flex Day

Materials: internet access, favorite recipes & cookbooks, culinary anthropology selection

→ EXPLORE

Compare recipes, watch any videos (in your Quick Links) that you find interesting and helpful, and think about how the ideas you've learned apply. What do you plan to make next week? Don't forget to get your shopping list to your teacher on time!

OR

→ READ, NARRATE, & DISCUSS

Current Events from Science News Explores or Food & Wine (in your Quick Links)

OR

Read more from your culinary anthropology selection.

WEEK 4 45m Food Science Lessons - Lesson 76

Preparation

Materials: Elements of Baking, Salt Fat Acid Heat, selected recipes

→ PREPARE

Read over the week's recipes. Consider how the recipes work and any substitutions/adjustments you plan to make. Use any extra time to journal a prelab narration in your notebook or gather ingredients and supplies.

• PLAN WEEKLY

- take a nature walk
- science free read

WEEK 4 45m Food Science Lessons - Lesson 77

The Science of Baking

Materials: Elements of Baking, selected recipe, indicated ingredients, and kitchen supplies

→ BAKING DAY!

Prepare your selected recipe. Record what you do and what you observe in your notebook.

WEEK 4 45m Food Science Lessons - Lesson 78

The Science of Baking

Materials: Elements of Baking, selected recipe

→ ANALYZE

- Review how the bake went, reread, and plan any adjustments that you

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want to try next week OR

- Select your next recipe. Read the recipe and any relevant information, such as the case study, if the recipe is in Elements of Baking. Add to the shopping list, as needed.

WEEK 4 45m Food Science Lessons - Lesson 79

Literature of Food Science & Culture

Materials: culinary anthropology selection

PREP: Read Teacher Tip

→ READ, NARRATE, & DISCUSS

Read at your own pace. Notice how the ingredients, tools, and customs surrounding food tell a story about the people. Think about how the ideas you have learned come together in the cultural development of the people. Use what you have learned to experiment and adopt some new recipes for your family.

★ TEACHER TIP

Students are scheduled to complete a citizen science project next week during which they will audit food waste. The project preparation begins in the next lesson. This project can be moved to any other week during the term based on interest or convenience.

WEEK 4 45m Food Science Lessons - Lesson 80

Materials: internet access

→ INTRO

Let's think about how we interact with food as citizens. Begin with a citizen science project about food waste. You will report on this project toward the end of the term.

→ PREPARE

Follow the link and explore the website to learn about the project. Scroll down to "Follow these steps to get started." Use today to prepare for the project. You will take a break from baking and cooking to complete the project next week. If you prefer, you may move this project to any other week this term.

∞ Website Link: Food Conservation

WEEK 5 45m Food Science Lessons - Lesson 81

Materials: internet access, kitchen scale

→ DATA COLLECTION

Complete the first Kitchen Waste Audit

∞ Website Link: Food Conservation

• PLAN WEEKLY

- ▣ take a nature walk
- ▣ science free read

WEEK 5 45m Food Science Lessons - Lesson 82

Materials: internet access, kitchen scale

→ DATA COLLECTION

Conduct Day 1 Food Waste Log

∞ Website Link: Food Conservation

WEEK 5 45m Food Science Lessons - Lesson 83

Materials: internet access, kitchen scale

→ DATA COLLECTION

Conduct Day 2 Food Waste Log

∞ Website Link: Food Conservation

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WEEK 5 45m Food Science Lessons - Lesson 84

Materials: internet access, kitchen scale

→ DATA COLLECTION

Conduct Day 3 Food Waste Log

∞ Website Link: Food Conservation

WEEK 5 45m Food Science Lessons - Lesson 85

Materials: internet access, kitchen scale

→ DATA COLLECTION

Conduct Day 4 Food Waste Log

∞ Website Link: Food Conservation

WEEK 6 45m Food Science Lessons - Lesson 86

Preparation

Materials: Elements of Baking, Salt, Fat, Acid, Heat, selected recipes

→ PREPARE

Read over the week's recipes. Consider how the recipes work and any substitutions/adjustments you plan to make. Use any extra time to journal a prelab narration in your notebook or gather ingredients and supplies.

• PLAN WEEKLY

- take a nature walk
- science free read

WEEK 6 45m Food Science Lessons - Lesson 87

The Science of Baking

Materials: Elements of Baking, selected recipe, indicated ingredients, and kitchen supplies

→ BAKING DAY!

Prepare your selected recipe. Record what you do and what you observe in your notebook.

WEEK 6 45m Food Science Lessons - Lesson 88

The Science of Baking

Materials: Elements of Baking, selected recipe

→ ANALYZE

- Review how the bake went, reread, and plan any adjustments that you want to try next week OR
- Select your next recipe. Read the recipe and any relevant information, such as the case study, if the recipe is in Elements of Baking. Add to the shopping list, as needed.

WEEK 6 45m Food Science Lessons - Lesson 89

Literature of Food Science & Culture

Materials: culinary anthropology selection

→ READ, NARRATE, & DISCUSS

Read at your own pace. Notice how the ingredients, tools, and customs surrounding food tell a story about the people. Think about how the ideas you have learned come together in the cultural development of the people. Use what you have learned to experiment and adopt some new recipes for your family.

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WEEK 6 ☐ 45m Food Science Lessons - Lesson 90

Flex Day

☐ Materials: internet access, favorite recipes & cookbooks, culinary anthropology selection

→ EXPLORE

Compare recipes, watch any videos (in your Quick Links) that you find interesting and helpful, and think about how the ideas you've learned apply. What do you plan to make next week? Don't forget to get your shopping list to your teacher on time!

OR

→ READ, NARRATE, & DISCUSS

Current Events from Science News Explores or Food & Wine (in your Quick Links)

OR

Read more from your culinary anthropology selection.

WEEK 7 ☐ 45m Food Science Lessons - Lesson 91

Preparation

☐ Materials: Elements of Baking, Salt, Fat, Acid, Heat, selected recipes

→ PREPARE

Read over the week's recipes. Consider how the recipes work and any substitutions/adjustments you plan to make. Use any extra time to journal a prelab narration in your notebook or gather ingredients and supplies.

• PLAN WEEKLY

- ☐ take a nature walk
- ☐ science free read

WEEK 7 ☐ 45m Food Science Lessons - Lesson 92

The Science of Baking

☐ Materials: Elements of Baking, selected recipe, indicated ingredients, and kitchen supplies

→ BAKING DAY!

Prepare your selected recipe. Record what you do and what you observe in your notebook.

WEEK 7 ☐ 45m Food Science Lessons - Lesson 93

The Science of Baking

☐ Materials: Elements of Baking, selected recipe

→ ANALYZE

- Review how the bake went, reread, and plan any adjustments that you want to try next week OR
- Select your next recipe. Read the recipe and any relevant information, such as the case study, if the recipe is in Elements of Baking. Add to the shopping list, as needed.

WEEK 7 ☐ 45m Food Science Lessons - Lesson 94

Literature of Food Science & Culture

☐ Materials: culinary anthropology selection

→ READ, NARRATE, & DISCUSS

Read at your own pace. Notice how the ingredients, tools, and customs

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surrounding food tell a story about the people. Think about how the ideas you have learned come together in the cultural development of the people. Use what you have learned to experiment and adopt some new recipes for your family.

WEEK 7 45m Food Science Lessons - Lesson 95

Flex Day

Materials: internet access, favorite recipes & cookbooks, culinary anthropology selection

→ EXPLORE

Compare recipes, watch any videos (in your Quick Links) that you find interesting and helpful, and think about how the ideas you've learned apply. What do you plan to make next week? Don't forget to get your shopping list to your teacher on time!

OR

→ READ, NARRATE, & DISCUSS

Current Events from Science News Explores or Food & Wine (in your Quick Links)

OR

Read more from your culinary anthropology selection.

WEEK 8 45m Food Science Lessons - Lesson 96

Preparation

Materials: Elements of Baking, Salt, Fat, Acid, Heat, selected recipes

→ PREPARE

Read over the week's recipes. Consider how the recipes work and any substitutions/adjustments you plan to make. Use any extra time to journal a prelab narration in your notebook or gather ingredients and supplies.

• PLAN WEEKLY

- take a nature walk
- science free read

WEEK 8 45m Food Science Lessons - Lesson 97

The Science of Baking

Materials: Elements of Baking, selected recipe, indicated ingredients, and kitchen supplies

→ BAKING DAY!

Prepare your selected recipe. Record what you do and what you observe in your notebook.

WEEK 8 45m Food Science Lessons - Lesson 98

The Science of Baking

Materials: Elements of Baking, selected recipe

→ ANALYZE

- Review how the bake went, reread, and plan any adjustments that you want to try next week OR
- Select your next recipe. Read the recipe and any relevant information, such as the case study, if the recipe is in Elements of Baking. Add to the shopping list, as needed.

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WEEK 8 45m Food Science Lessons - Lesson 99

Literature of Food Science & Culture

Materials: culinary anthropology selection

→ READ, NARRATE, & DISCUSS

Read at your own pace. Notice how the ingredients, tools, and customs surrounding food tell a story about the people. Think about how the ideas you have learned come together in the cultural development of the people. Use what you have learned to experiment and adopt some new recipes for your family.

WEEK 8 45m Food Science Lessons - Lesson 100

Flex Day

Materials: internet access, favorite recipes & cookbooks, culinary anthropology selection

→ EXPLORE

Compare recipes, watch any videos (in your Quick Links) that you find interesting and helpful, and think about how the ideas you've learned apply. What do you plan to make next week? Don't forget to get your shopping list to your teacher on time!

OR

→ READ, NARRATE, & DISCUSS

Current Events from Science News Explores or Food & Wine (in your Quick Links)

OR

Read more from your culinary anthropology selection.

WEEK 9 45m Food Science Lessons - Lesson 101

Preparation

Materials: Elements of Baking, Salt, Fat, Acid, Heat, selected recipes

→ PREPARE

Read over the week's recipes. Consider how the recipes work and any substitutions/adjustments you plan to make. Use any extra time to journal a prelab narration in your notebook or gather ingredients and supplies.

• PLAN WEEKLY

- take a nature walk
- science free read

WEEK 9 45m Food Science Lessons - Lesson 102

The Science of Baking

Materials: Elements of Baking, selected recipe, indicated ingredients, and kitchen supplies

→ BAKING DAY!

Prepare your selected recipe. Record what you do and what you observe in your notebook.

WEEK 9 45m Food Science Lessons - Lesson 103

The Science of Baking

Materials: Elements of Baking, selected recipe

→ ANALYZE

- Review how the bake went, reread, and plan any adjustments that you

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want to try next week OR

- Select your next recipe. Read the recipe and any relevant information, such as the case study, if the recipe is in Elements of Baking. Add to the shopping list, as needed.

WEEK 9 45m Food Science Lessons - Lesson 104

Literature of Food Science & Culture

Materials: culinary anthropology selection

→ READ, NARRATE, & DISCUSS

Read at your own pace. Notice how the ingredients, tools, and customs surrounding food tell a story about the people. Think about how the ideas you have learned come together in the cultural development of the people. Use what you have learned to experiment and adopt some new recipes for your family.

WEEK 9 45m Food Science Lessons - Lesson 105

Flex Day

Materials: internet access, favorite recipes & cookbooks, culinary anthropology selection

→ EXPLORE

Compare recipes, watch any videos (in your Quick Links) that you find interesting and helpful, and think about how the ideas you've learned apply. What do you plan to make next week? Don't forget to get your shopping list to your teacher on time!

OR

→ READ, NARRATE, & DISCUSS

Current Events from Science News Explores or Food & Wine (in your Quick Links)

OR

Read more from your culinary anthropology selection.

WEEK 10 45m Food Science Lessons - Lesson 106

Preparation

Materials: Elements of Baking, Salt, Fat, Acid, Heat, selected recipes

→ PREPARE

Read over the week's recipes. Consider how the recipes work and any substitutions/adjustments you plan to make. Use any extra time to journal a prelab narration in your notebook or gather ingredients and supplies.

• PLAN WEEKLY

- ☐ take a nature walk
- ☐ science free read

WEEK 10 45m Food Science Lessons - Lesson 107

The Science of Baking

Materials: Elements of Baking, selected recipe, indicated ingredients, and kitchen supplies

→ BAKING DAY!

Prepare your selected recipe. Record what you do and what you observe in your notebook.

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WEEK 10 45m Food Science Lessons - Lesson 108

The Science of Baking

Materials: Elements of Baking, selected recipe

→ ANALYZE

- Review how the bake went, reread, and plan any adjustments that you want to try next week OR
- Select your next recipe. Read the recipe and any relevant information, such as the case study, if the recipe is in Elements of Baking. Add to the shopping list, as needed.

WEEK 10 45m Food Science Lessons - Lesson 109

Literature of Food Science & Culture

Materials: culinary anthropology selection

→ READ, NARRATE, & DISCUSS

Read at your own pace. Notice how the ingredients, tools, and customs surrounding food tell a story about the people. Think about how the ideas you have learned come together in the cultural development of the people. Use what you have learned to experiment and adopt some new recipes for your family.

WEEK 10 45m Food Science Lessons - Lesson 110

Flex Day

Materials: internet access, favorite recipes & cookbooks, culinary anthropology selection

→ EXPLORE

Compare recipes, watch any videos (in your Quick Links) that you find interesting and helpful, and think about how the ideas you've learned apply. What do you plan to make next week? Don't forget to get your shopping list to your teacher on time!

OR

→ READ, NARRATE, & DISCUSS

Current Events from Science News Explores or Food & Wine (in your Quick Links)

OR

Read more from your culinary anthropology selection.

WEEK 11 45m Food Science Lessons - Lesson 111

Preparation

Materials: Elements of Baking, Salt, Fat, Acid, Heat, selected recipes

→ PREPARE

Read over the week's recipes. Consider how the recipes work and any substitutions/adjustments you plan to make. Use any extra time to journal a prelab narration in your notebook or gather ingredients and supplies.

• PLAN WEEKLY

- ☐ take a nature walk
- ☐ science free read

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WEEK 11 45m Food Science Lessons - Lesson 112

The Science of Baking

Materials: Elements of Baking, selected recipe, indicated ingredients, and kitchen supplies

→ BAKING DAY!

Prepare your selected recipe. Record what you do and what you observe in your notebook.

WEEK 11 45m Food Science Lessons - Lesson 113

The Science of Baking

Materials: Elements of Baking, selected recipe

→ ANALYZE

- Review how the bake went and consider what you learned.

WEEK 11 45m Food Science Lessons - Lesson 114

Present What You've Learned

→ PREPARE

For one of your exam questions, you will be asked to give an informal presentation to your teacher on your citizen science project. This can be an oral presentation (e.g., a poster presentation, PowerPoint slides, etc) or something you've prepared (e.g., an essay, Canva brochure, or graphic, painting, etc). Use this time to gather your thoughts and to think through how you want to do that. You can be as creative as you would like.

WEEK 11 45m Food Science Lessons - Lesson 115

Present What You've Learned

→ PREPARE

For one of your exam questions, you will be asked to give an informal presentation to your teacher on your citizen science project. This can be an oral presentation (e.g., a poster presentation, PowerPoint slides, etc) or something you've prepared (e.g., an essay, Canva brochure, or graphic, painting, etc). Use this time to gather your thoughts and to think through how you want to do that. You can be as creative as you would like.

WEEK 12 45m Food Science Lessons - Lesson 116

Exam Week

→ EXAMS

Answer question(s) related to the course.

WEEK 12 45m Food Science Lessons - Lesson 117

Exam Week

→ EXAMS

Answer question(s) related to the course.

WEEK 12 45m Food Science Lessons - Lesson 118

Exam Week

→ EXAMS

Answer question(s) related to the course.

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WEEK 12 45m Food Science Lessons - Lesson 119 <i>Exam Week</i> → EXAMS Answer question(s) related to the course.	
WEEK 12 45m Food Science Lessons - Lesson 120 <i>Exam Week</i> → EXAMS Answer question(s) related to the course.	

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WEEK 1 45m Food Science Lessons - Lesson 121

Preparation

Materials: internet access, favorite recipes & cookbooks

→ INTRO

Schedule your baking day and dinner night, but otherwise, spend time comparing recipes, watching any videos (below and in your Quick Links) that you find interesting and helpful, and thinking about how the ideas apply - don't forget to get your shopping list to your teacher on time!

∞ Video Link: America's Test Kitchen has an extensive video library on YouTube (the playlist included in term 1 a few times is called What's Eating Dan)

∞ Video Link: The Loopy Whisk is by the baker and chemist who wrote The Elements of Baking

∞ Video Link: Samin Nosrat Cooking features the chef who wrote Salt, Fat, Acid, Heat

• PLAN WEEKLY

- take a nature walk
- science free read

WEEK 1 45m Food Science Lessons - Lesson 122

The Science of Baking

Materials: Elements of Baking

→ PLAN

Consider what lessons you would like to learn this term; what recipes you would like to experiment with (in this book or from any other source); what ingredients you would like to better understand. Set a goal for this term and plan using the skills you have learned.

WEEK 1 45m Food Science Lessons - Lesson 123

The Science of Cooking

Materials: Salt, Fat, Acid, Heat

→ PLAN

Consider what cooking lessons you would like to learn this term; what recipes you would like to experiment with (in this book or from any other source); what ingredients you would like to better understand. Set a goal for this term and plan using the skills you have learned.

WEEK 1 45m Food Science Lessons - Lesson 124

Literature of Food Science & Culture

Materials: Culinary Reactions

→ READ, NARRATE, & DISCUSS

About 1 chapter. You won't have time to finish this book in this one lesson slot for the term. You could choose to read more during your flex day, during your weekly free read time, or on your own at the end of term. If you choose not to finish the book, that is fine, too.

WEEK 1 45m Food Science Lessons - Lesson 125

Flex Day

Materials: internet access, favorite recipes & cookbooks, Culinary Reactions

→ EXPLORE

Compare recipes, watch any videos (in your Quick Links) that you find interesting and helpful, and think about how the ideas you've learned apply. What do you plan to make next week? Don't forget to get your

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shopping list to your teacher on time!

OR

→ READ, NARRATE, & DISCUSS

Current Events from Science News Explores or Food & Wine (in your Quick Links)

OR

Read another chapter in Culinary Reactions, as you may not have time to finish this book in a single term.

WEEK 2 45m Food Science Lessons - Lesson 126

Preparation

Materials: Elements of Baking, Salt, Fat, Acid, Heat, selected recipes

→ PREPARE

Read over the week's recipes. Consider how the recipes work and any substitutions/adjustments you plan to make. Use any extra time to journal a prelab narration in your notebook or gather ingredients and supplies.

• PLAN WEEKLY

- take a nature walk
- science free read

WEEK 2 45m Food Science Lessons - Lesson 127

The Science of Baking

Materials: Elements of Baking, selected recipe, indicated ingredients, and kitchen supplies

→ BAKING DAY!

Prepare your selected recipe. Record what you do and what you observe in your notebook.

WEEK 2 45m Food Science Lessons - Lesson 128

The Science of Baking

Materials: Elements of Baking, selected recipe

→ ANALYZE

- Review how the bake went, reread, and plan any adjustments that you want to try next week OR
- Select your next recipe. Read the recipe and any relevant information, such as the case study, if the recipe is in Elements of Baking. Add to the shopping list, as needed.

WEEK 2 45m Food Science Lessons - Lesson 129

Literature of Food Science & Culture

Materials: Culinary Reactions

→ READ, NARRATE, & DISCUSS

About 1 chapter

WEEK 2 45m Food Science Lessons - Lesson 130

Flex Day

Materials: internet access, favorite recipes & cookbooks, Culinary Reactions

→ EXPLORE

Compare recipes, watch any videos (in your Quick Links) that you find



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interesting and helpful, and think about how the ideas you've learned apply. What do you plan to make next week? Don't forget to get your shopping list to your teacher on time!

OR

→ READ, NARRATE, & DISCUSS

Current Events from Science News Explores or Food & Wine (in your Quick Links)

OR

Read another chapter in Culinary Reactions, as you may not have time to finish this book in a single term.

WEEK 3 ☐ 45m Food Science Lessons - Lesson 131

Preparation

☐ Materials: Elements of Baking, Salt, Fat, Acid, Heat, selected recipes

→ PREPARE

Read over the week's recipes. Consider how the recipes work and any substitutions/adjustments you plan to make. Use any extra time to journal a prelab narration in your notebook or gather ingredients and supplies.

• PLAN WEEKLY

- ☐ take a nature walk
- ☐ science free read

WEEK 3 ☐ 45m Food Science Lessons - Lesson 132

The Science of Baking

☐ Materials: Elements of Baking, selected recipe, indicated ingredients, and kitchen supplies

→ BAKING DAY!

Prepare your selected recipe. Record what you do and what you observe in your notebook.

WEEK 3 ☐ 45m Food Science Lessons - Lesson 133

The Science of Baking

☐ Materials: Elements of Baking, selected recipe

→ ANALYZE

- Review how the bake went, reread, plan any adjustments that you want to try in 2 weeks (there is a project scheduled next week) OR
- Select your next recipe. Read the recipe and any relevant information, such as the case study, if the recipe is in Elements of Baking. Add to the shopping list, as needed.

WEEK 3 ☐ 45m Food Science Lessons - Lesson 134

Literature of Food Science & Culture

☐ Materials: Culinary Reactions

→ READ, NARRATE, & DISCUSS

About 1 chapter

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WEEK 3 ☐ 45m Food Science Lessons - Lesson 135

Flex Day

☐ Materials: internet access, favorite recipes & cookbooks, Culinary Reactions

→ EXPLORE

Compare recipes, watch any videos (in your Quick Links) that you find interesting and helpful, and think about how the ideas you've learned apply. What do you plan to make next week? Don't forget to get your shopping list to your teacher on time!

OR

→ READ, NARRATE, & DISCUSS

Current Events from Science News Explores or Food & Wine (in your Quick Links)

OR

Read another chapter in Culinary Reactions, as you may not have time to finish this book in a single term.

WEEK 4 ☐ 45m Food Science Lessons - Lesson 136

Preparation

☐ Materials: Salt, Fat, Acid, Heat, selected recipe

→ PREPARE

Read over the week's dinner recipe. Consider how the recipe works and any substitutions/adjustments you plan to make. Use any extra time to journal a prelab narration in your notebook, gather ingredients and supplies, or get started on your research project.

• PLAN WEEKLY

- ☐ take a nature walk
- ☐ science free read

WEEK 4 ☐ 45m Food Science Lessons - Lesson 137

☐ Materials: internet or library access

→ INTRO

Last term, we evaluated food waste in our community. This is one important aspect of how we interact with food as citizens. Another is being aware of where our food comes from and how the production and process of getting that food to our plates impacts the world around us. Watch the linked video to begin thinking about the problem. While there is no one-size-fits-all approach, we want to make decisions that do the most good and the least harm. You will have three periods for this research and will prepare a composition about your findings.
∞ Video Link: What is the true price of the food we eat?

→ RESEARCH

Select 3 ingredients from your last dinner recipe. For each ingredient, try to answer these questions:

- How is this ingredient produced? Is it grown on farms? Is it further processed in a factory?
- Where is it produced?
- How is it packaged and distributed?
- What kind of waste is generated from production, packaging, and distribution?
- How does the whole process positively and negatively affect people and the environment?
- What systems in society support the use of this energy source (e.g., economic, transportation, etc.)?

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Term 3

WEEK 4 45m Food Science Lessons - Lesson 138

Materials: internet or library access

→ RESEARCH

Continue researching food production.

WEEK 4 45m Food Science Lessons - Lesson 139

Materials: internet or library access

→ RESEARCH

Complete your research. Use any remaining time to begin thinking about your composition. Effective communication of complex ideas is a major challenge in science, so think about how you communicate best and plan your composition thoughtfully! You could compose an expository essay, a PowerPoint or poster presentation, a graphic display using Canva or watercolor, or any other form of expression that allows you to communicate what you have learned effectively.

WEEK 4 45m Food Science Lessons - Lesson 140

Literature of Food Science & Culture

Materials: Culinary Reactions

→ READ, NARRATE, & DISCUSS

About 1 chapter

WEEK 5 45m Food Science Lessons - Lesson 141

Preparation

Materials: Elements of Baking, Salt Fat Acid Heat, selected recipes

→ PREPARE

Read over the week's recipes. Consider how the recipes work and any substitutions/adjustments you plan to make. Use any extra time to journal a prelab narration in your notebook or gather ingredients and supplies.

• PLAN WEEKLY

- take a nature walk
- science free read

WEEK 5 45m Food Science Lessons - Lesson 142

The Science of Baking

Materials: Elements of Baking, selected recipe, indicated ingredients, and kitchen supplies

→ BAKING DAY!

Prepare your selected recipe. Record what you do and what you observe in your notebook.

WEEK 5 45m Food Science Lessons - Lesson 143

The Science of Baking

Materials: Elements of Baking, selected recipe

→ ANALYZE

- Review how the bake went, reread, and plan any adjustments that you want to try next week OR
- Select your next recipe. Read the recipe and any relevant information, such as the case study, if the recipe is in Elements of Baking. Add to the shopping list, as needed.

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Term 3

WEEK 5 45m Food Science Lessons - Lesson 144

Literature of Food Science & Culture

Materials: Culinary Reactions

→ READ, NARRATE, & DISCUSS

About 1 chapter

WEEK 5 45m Food Science Lessons - Lesson 145

Flex Day

Materials: internet access, favorite recipes & cookbooks, Culinary Reactions

→ EXPLORE

Compare recipes, watch any videos (in your Quick Links) that you find interesting and helpful, and think about how the ideas you've learned apply. What do you plan to make next week? Don't forget to get your shopping list to your teacher on time!

OR

→ READ, NARRATE, & DISCUSS

Current Events from Science News Explores or Food & Wine (in your Quick Links)

OR

Read another chapter in Culinary Reactions, as you may not have time to finish this book in a single term.

WEEK 6 45m Food Science Lessons - Lesson 146

Preparation

Materials: Elements of Baking, Salt Fat Acid Heat, selected recipes

→ PREPARE

Read over the week's recipes. Consider how the recipes work and any substitutions/adjustments you plan to make. Use any extra time to journal a prelab narration in your notebook or gather ingredients and supplies.

• PLAN WEEKLY

- take a nature walk
- science free read

WEEK 6 45m Food Science Lessons - Lesson 147

The Science of Baking

Materials: Elements of Baking, selected recipe, indicated ingredients, and kitchen supplies

→ BAKING DAY!

Prepare your selected recipe. Record what you do and what you observe in your notebook.

WEEK 6 45m Food Science Lessons - Lesson 148

The Science of Baking

Materials: Elements of Baking, selected recipe

→ ANALYZE

- Review how the bake went, reread, and plan any adjustments that you want to try next week OR

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- Select your next recipe. Read the recipe and any relevant information, such as the case study, if the recipe is in Elements of Baking. Add to the shopping list, as needed.

WEEK 6 45m Food Science Lessons - Lesson 149

Literature of Food Science & Culture

Materials: Culinary Reactions

→ **READ, NARRATE, & DISCUSS**
About 1 chapter

WEEK 6 45m Food Science Lessons - Lesson 150

Flex Day

Materials: internet access, favorite recipes & cookbooks, Culinary Reactions

→ EXPLORE

Compare recipes, watch any videos (in your Quick Links) that you find interesting and helpful, and think about how the ideas you've learned apply. What do you plan to make next week? Don't forget to get your shopping list to your teacher on time!

OR

→ **READ, NARRATE, & DISCUSS**

Current Events from Science News Explores or Food & Wine (in your Quick Links)

OR

Read another chapter in Culinary Reactions, as you may not have time to finish this book in a single term.

WEEK 7 45m Food Science Lessons - Lesson 151

Preparation

Materials: Elements of Baking, Salt, Fat, Acid, Heat, selected recipes

→ **PREPARE**

Read over the week's recipes. Consider how the recipes work and any substitutions/adjustments you plan to make. Use any extra time to journal a prelab narration in your notebook or gather ingredients and supplies.

• **PLAN WEEKLY**

- ☐ take a nature walk
- ☐ science free read

WEEK 7 45m Food Science Lessons - Lesson 152

The Science of Baking

Materials: Elements of Baking, selected recipe, indicated ingredients, and kitchen supplies

→ **BAKING DAY!**

Prepare your selected recipe. Record what you do and what you observe in your notebook.

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WEEK 7 ☐ 45m Food Science Lessons - Lesson 153

The Science of Baking

☐ Materials: Elements of Baking, selected recipe

→ ANALYZE

- Review how the bake went, reread, and plan any adjustments that you want to try next week OR
- Select your next recipe. Read the recipe and any relevant information, such as the case study, if the recipe is in Elements of Baking. Add to the shopping list, as needed.

WEEK 7 ☐ 45m Food Science Lessons - Lesson 154

Literature of Food Science & Culture

☐ Materials: Culinary Reactions

→ READ, NARRATE, & DISCUSS

About 1 chapter

WEEK 7 ☐ 45m Food Science Lessons - Lesson 155

Flex Day

☐ Materials: internet access, favorite recipes & cookbooks, Culinary Reactions

→ EXPLORE

Compare recipes, watch any videos (in your Quick Links) that you find interesting and helpful, and think about how the ideas you've learned apply. What do you plan to make next week? Don't forget to get your shopping list to your teacher on time!

OR

→ READ, NARRATE, & DISCUSS

Current Events from Science News Explores or Food & Wine (in your Quick Links)

OR

Read another chapter in Culinary Reactions, as you may not have time to finish this book in a single term.

WEEK 8 ☐ 45m Food Science Lessons - Lesson 156

Preparation

☐ Materials: Elements of Baking, Salt Fat Acid Heat, selected recipes

→ PREPARE

Read over the week's recipes. Consider how the recipes work and any substitutions/adjustments you plan to make. Use any extra time to journal a prelab narration in your notebook or gather ingredients and supplies.

• PLAN WEEKLY

- ☐ take a nature walk
- ☐ science free read

WEEK 8 ☐ 45m Food Science Lessons - Lesson 157

The Science of Baking

☐ Materials: Elements of Baking, selected recipe, indicated ingredients, and kitchen supplies

→ BAKING DAY!

Prepare your selected recipe. Record what you do and what you observe

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in your notebook.

WEEK 8 45m Food Science Lessons - Lesson 158

The Science of Baking

Materials: Elements of Baking, selected recipe

→ ANALYZE

- Review how the bake went, reread, and plan any adjustments that you want to try next week OR
- Select your next recipe. Read the recipe and any relevant information, such as the case study, if the recipe is in Elements of Baking. Add to the shopping list, as needed.

WEEK 8 45m Food Science Lessons - Lesson 159

Literature of Food Science & Culture

Materials: Culinary Reactions

→ READ, NARRATE, & DISCUSS

About 1 chapter

WEEK 8 45m Food Science Lessons - Lesson 160

Flex Day

Materials: internet access, favorite recipes & cookbooks, Culinary Reactions

→ EXPLORE

Compare recipes, watch any videos (in your Quick Links) that you find interesting and helpful, and think about how the ideas you've learned apply. What do you plan to make next week? Don't forget to get your shopping list to your teacher on time!

OR

→ READ, NARRATE, & DISCUSS

Current Events from Science News Explores or Food & Wine (in your Quick Links)

OR

Read another chapter in Culinary Reactions, as you may not have time to finish this book in a single term.

WEEK 9 45m Food Science Lessons - Lesson 161

Preparation

Materials: Elements of Baking, Salt, Fat, Acid, Heat, selected recipes

→ PREPARE

Read over the week's recipes. Consider how the recipes work and any substitutions/adjustments you plan to make. Use any extra time to journal a prelab narration in your notebook or gather ingredients and supplies.

• PLAN WEEKLY

- ☐ take a nature walk
- ☐ science free read

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WEEK 9 45m Food Science Lessons - Lesson 162

The Science of Baking

Materials: Elements of Baking, selected recipe, indicated ingredients, and kitchen supplies

→ BAKING DAY!

Prepare your selected recipe. Record what you do and what you observe in your notebook.

WEEK 9 45m Food Science Lessons - Lesson 163

The Science of Baking

Materials: Elements of Baking, selected recipe

→ ANALYZE

- Review how the bake went, reread, and plan any adjustments that you want to try next week OR
- Select your next recipe. Read the recipe and any relevant information, such as the case study, if the recipe is in Elements of Baking. Add to the shopping list, as needed.

WEEK 9 45m Food Science Lessons - Lesson 164

Literature of Food Science & Culture

Materials: Culinary Reactions

→ READ, NARRATE, & DISCUSS

About 1 chapter

WEEK 9 45m Food Science Lessons - Lesson 165

Flex Day

Materials: internet access, favorite recipes & cookbooks, Culinary Reactions

→ EXPLORE

Compare recipes, watch any videos (in your Quick Links) that you find interesting and helpful, and think about how the ideas you've learned apply. What do you plan to make next week? Don't forget to get your shopping list to your teacher on time!

OR

→ READ, NARRATE, & DISCUSS

Current Events from Science News Explores or Food & Wine (in your Quick Links)

OR

Read another chapter in Culinary Reactions, as you may not have time to finish this book in a single term.

WEEK 10 45m Food Science Lessons - Lesson 166

Preparation

Materials: Elements of Baking, Salt Fat Acid Heat, selected recipes

→ PREPARE

Read over the week's recipes. Consider how the recipes work and any

• PLAN WEEKLY

- take a nature walk
- science free read

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substitutions/adjustments you plan to make. Use any extra time to journal a prelab narration in your notebook or gather ingredients and supplies.

WEEK 10 45m Food Science Lessons - Lesson 167

The Science of Baking

Materials: Elements of Baking, selected recipe, indicated ingredients, and kitchen supplies

→ BAKING DAY!

Prepare your selected recipe. Record what you do and what you observe in your notebook.

WEEK 10 45m Food Science Lessons - Lesson 168

The Science of Baking

Materials: Elements of Baking, selected recipe

→ ANALYZE

- Review how the bake went, reread, and plan any adjustments that you want to try next week OR
- Select your next recipe. Read the recipe and any relevant information, such as the case study, if the recipe is in Elements of Baking. Add to the shopping list, as needed.

WEEK 10 45m Food Science Lessons - Lesson 169

Literature of Food Science & Culture

Materials: Culinary Reactions

→ READ, NARRATE, & DISCUSS

About 1 chapter

WEEK 10 45m Food Science Lessons - Lesson 170

Flex Day

Materials: internet access, favorite recipes & cookbooks, Culinary Reactions

→ EXPLORE

Compare recipes, watch any videos (in your Quick Links) that you find interesting and helpful, and think about how the ideas you've learned apply. What do you plan to make next week? Don't forget to get your shopping list to your teacher on time!

OR

→ READ, NARRATE, & DISCUSS

Current Events from Science News Explores or Food & Wine (in your Quick Links)

OR

Read another chapter in Culinary Reactions, as you may not have time to finish this book in a single term.

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WEEK 11 45m Food Science Lessons - Lesson 171

Preparation

Materials: Elements of Baking, Salt, Fat, Acid, Heat, selected recipes

→ PREPARE

Read over the week's recipes. Consider how the recipes work and any substitutions/adjustments you plan to make. Use any extra time to journal a prelab narration in your notebook or gather ingredients and supplies.

• PLAN WEEKLY

- take a nature walk
- science free read

WEEK 11 45m Food Science Lessons - Lesson 172

The Science of Baking

Materials: Elements of Baking, selected recipe, indicated ingredients, and kitchen supplies

→ BAKING DAY!

Prepare your selected recipe. Record what you do and what you observe in your notebook.

WEEK 11 45m Food Science Lessons - Lesson 173

The Science of Baking

Materials: Elements of Baking, selected recipe

→ ANALYZE

- Review how the bake went and consider what you learned.

WEEK 11 45m Food Science Lessons - Lesson 174

Present What You've Learned

→ PREPARE

For one of your exam questions, you will be asked to give an informal presentation to your teacher on your citizen science research. This can be an oral presentation (e.g., a poster presentation, PowerPoint slides, etc.) or something you've prepared (e.g., an essay, Canva brochure, or graphic, painting, etc.). Use this time to gather your thoughts and to think through how you want to do that. You can be as creative as you would like.

WEEK 11 45m Food Science Lessons - Lesson 175

Present What You've Learned

→ PREPARE

For one of your exam questions, you will be asked to give an informal presentation to your teacher on your citizen science research. This can be an oral presentation (e.g., a poster presentation, PowerPoint slides, etc.) or something you've prepared (e.g., an essay, Canva brochure or graphic, painting, etc.). Use this time to gather your thoughts and to think through how you want to do that. You can be as creative as you would like.

WEEK 12 45m Food Science Lessons - Lesson 176

Exam Week

→ EXAMS

Answer question(s) related to the course.

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WEEK 12 45m Food Science Lessons - Lesson 177 Exam Week → EXAMS Answer question(s) related to the course.	
WEEK 12 45m Food Science Lessons - Lesson 178 Exam Week → EXAMS Answer question(s) related to the course.	
WEEK 12 45m Food Science Lessons - Lesson 179 Exam Week → EXAMS Answer question(s) related to the course.	
WEEK 12 45m Food Science Lessons - Lesson 180 Exam Week → EXAMS Answer question(s) related to the course.	

Food Science Lessons

Examination

Term 1

- Salt Fat Acid Heat: Tell something you learned about the chemistry of salt OR fat (lipids) OR acid, and the observations you made about this ingredient in the kitchen.
- Salt, Fat, Acid, Heat: Tell something you learned about what heat is and what it does physically/chemically. Also, tell about the observations you made about heat in the kitchen.
- Elements of Baking: Tell about one 'element of baking' that you read about this term, including what properties it contributes to a bake.
- Tell about a meal you prepared for 'dinner night' and explain how it was made. You may refer to the recipe to help, but you should be able to explain HOW to prepare the food and tell about how salt/acid/fat/heat contribute to the success of the dish. Share your lab notebook with your teacher.
- Give an informal presentation to your teacher on something you have learned about this term, such as how a particular recipe or ingredient works.

Term 2

- Share your lab notebook with your teacher and explain one of your bakes. What worked/didn't? How did ingredient substitutions affect the product?
- Culinary Anthropology selection: Tell what you learned about food in your selected culture. How are salt, fat, acid, and heat used by your selected culture?
- Tell something that stood out to you about a chosen recipe, current event, OR your culinary anthropology selection.
- Tell about a meal you prepared for 'dinner night' and explain how it was made. You may refer to the recipe to help, but you should be able to explain HOW to prepare the food and tell about how salt/acid/fat/heat contribute to the success of the dish. Share your notebook with your teacher.
- Give an informal presentation to your teacher on your citizen science project.

Term 3

- Share your lab notebook with your teacher and explain one of your bakes. What worked/didn't? How did ingredient substitutions affect the product? How does notebooking impact your learning experience?
- Explain an idea that you learned about in Culinary Reactions.
- Tell something that stood out to you about a chosen recipe, current event, OR Culinary Reactions.
- Tell about a meal you prepared for 'dinner night' and explain how it was made. You may refer to the recipe to help, but you should be able to explain HOW to prepare the food and tell about how salt/acid/fat/heat contribute to the success of the dish. Share your notebook with your teacher.
- Give an informal presentation to your teacher on your citizen science research.